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Coplin et al.

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(54) **LIQUID DISPENSING SYSTEM WITH
PRESSURIZED AIR OPERATED LIQUID
CONTROL PISTONS**

B05B 12/02; B05B 1/34; B05B 1/169;
B67D 2210/0016; F16K 31/122; F16K
31/1221; F16K 31/1245; F16K 31/124;
F16K 31/40; F16K 31/406; F16K 31/36;
(Continued)

(71) Applicant: **Spraying Systems Co.**, Wheaton, IL
(US)

(72) Inventors: **Thomas L. Coplin**, Lake in the Hill, IL
(US); **Edward Camic**, Wheaton, IL
(US)

(73) Assignee: **Spraying Systems Co.**, Wheaton, IL
(US)

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9, 2020.

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B05B 1/26 (2006.01)
B05B 1/16 (2006.01)
B05B 1/30 (2006.01)
B05B 1/34 (2006.01)

(Continued)

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CPC **B05B 1/265** (2013.01); **B05B 1/169**
(2013.01); **B05B 1/3013** (2013.01); **B05B**
1/3053 (2013.01); **B05B 1/34** (2013.01); **B05B**
12/02 (2013.01); **F16K 31/1221** (2013.01)

(58) **Field of Classification Search**
CPC B05B 1/265; B05B 1/3013; B05B 1/3053;

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Primary Examiner — Arthur O. Hall

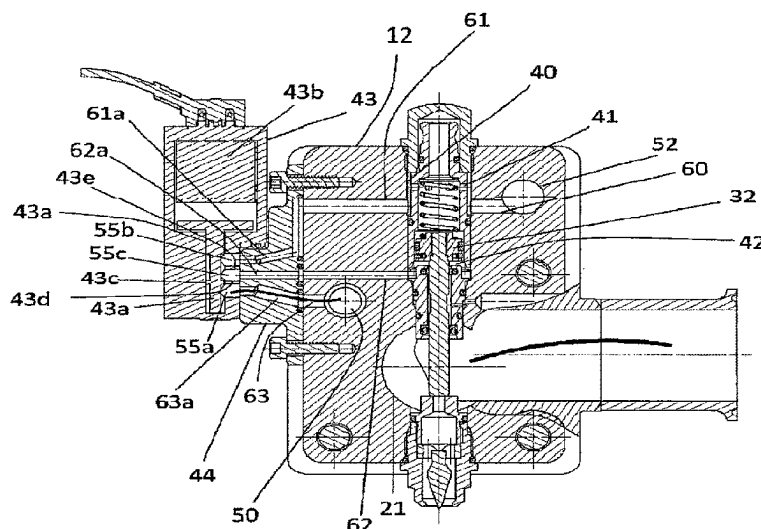
Assistant Examiner — Juan C Barrera

(74) *Attorney, Agent, or Firm* — Leydig, Voit & Mayer,
Ltd.

(57) **ABSTRACT**

A modular liquid distribution system in which each module
has a module body, a spray nozzle, and a piston for con-
trolling the dispensing of liquid from the nozzle. Each
module has a pneumatically operated system for moving the
piston to an open position while facilitating quicker return
movement to a closed position, enabling the dispensing of
precisely controlled small droplet sized quantities of highly
viscous liquids.

13 Claims, 12 Drawing Sheets



- * cited by examiner

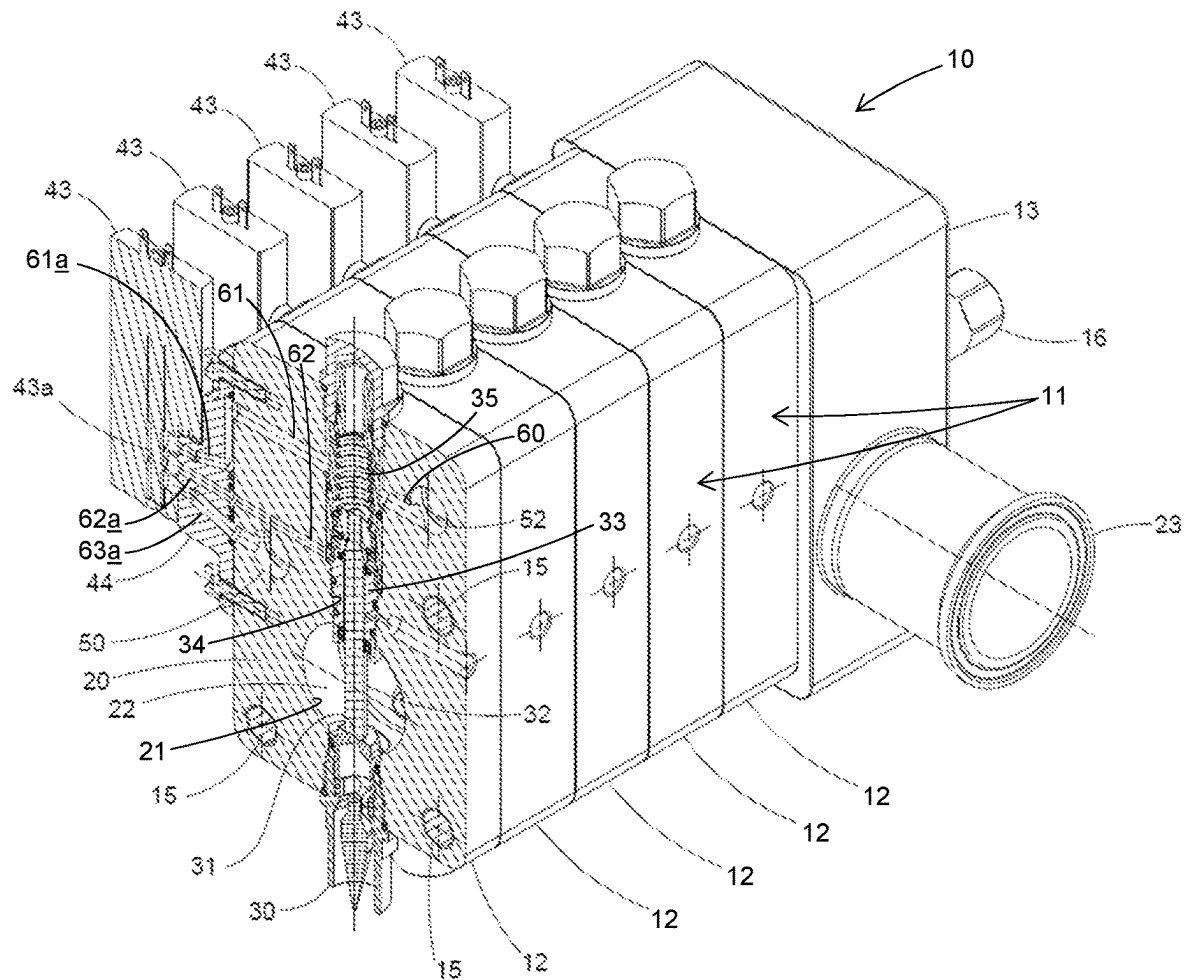


FIG. 1

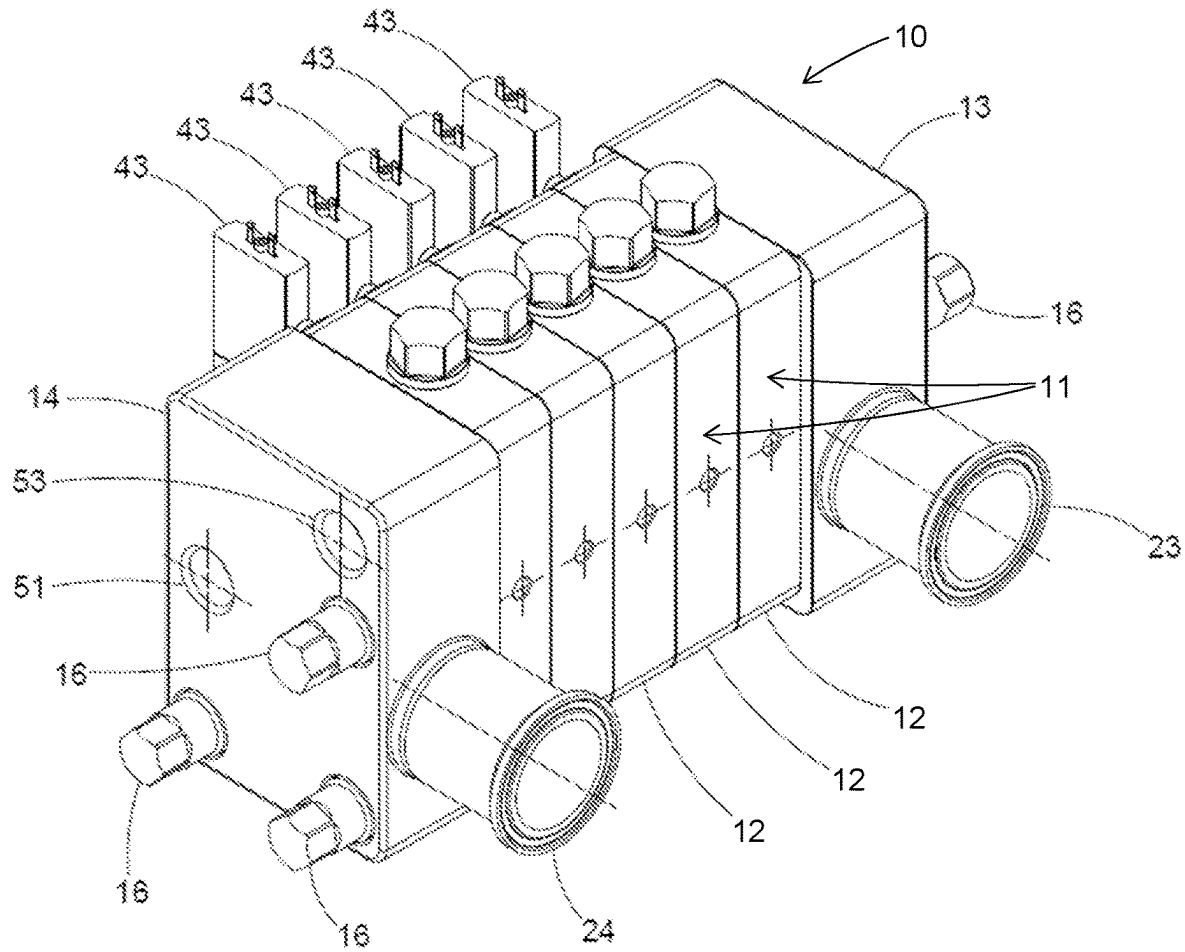


FIG. 2

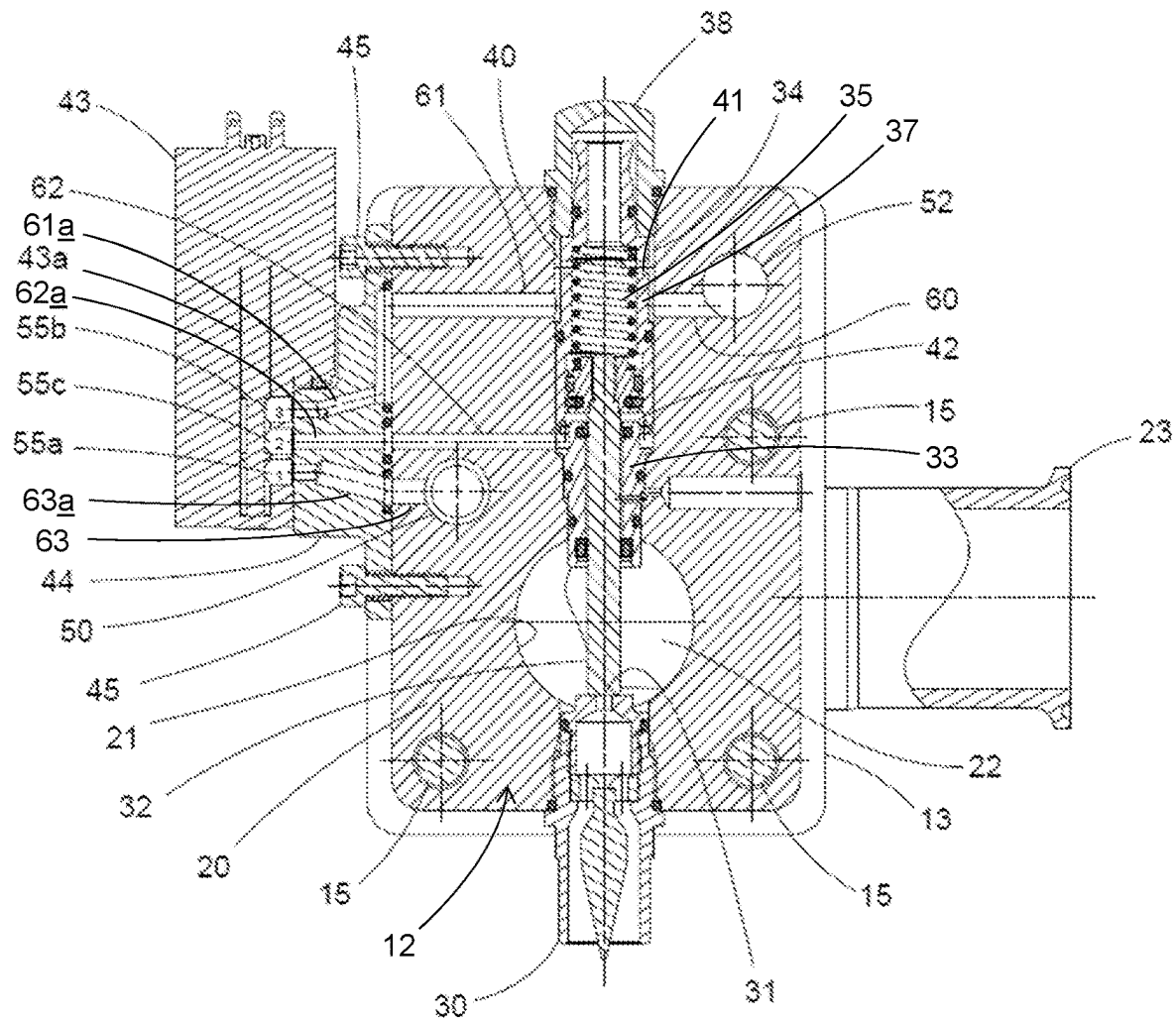


FIG. 3

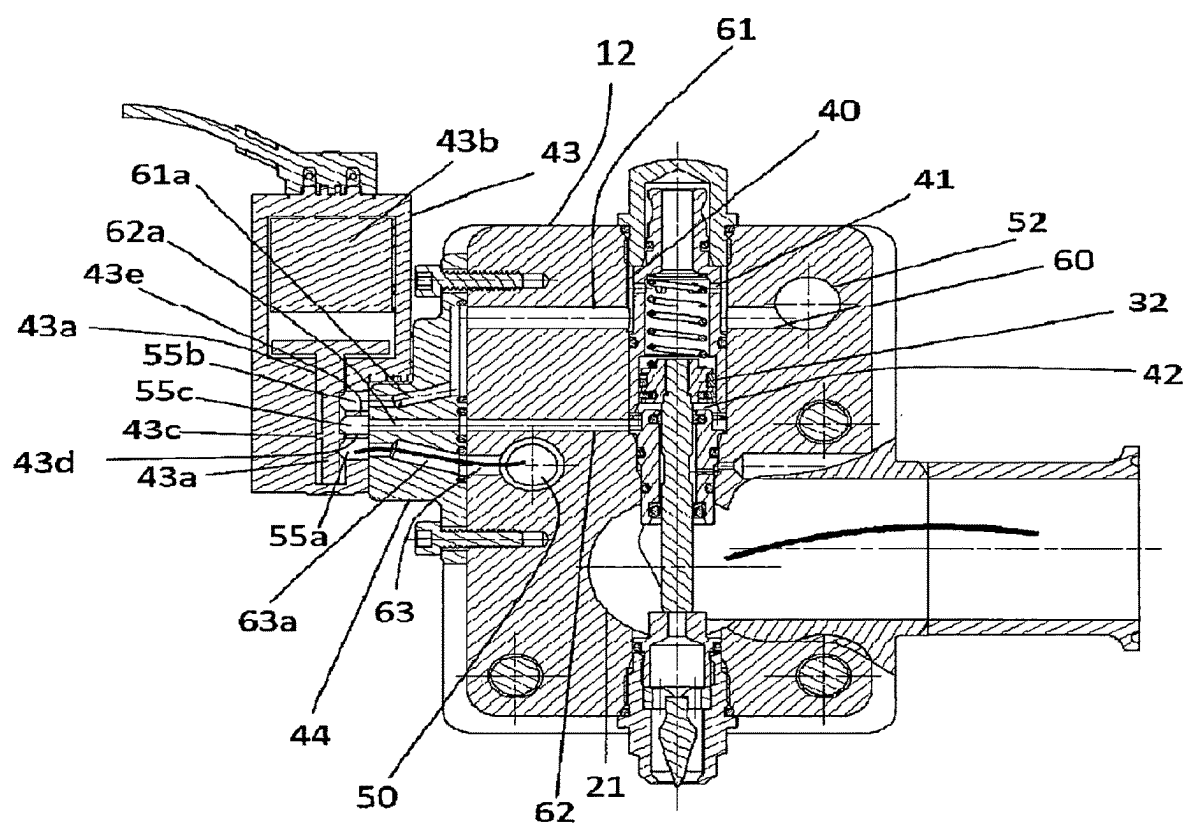


FIG. 3A

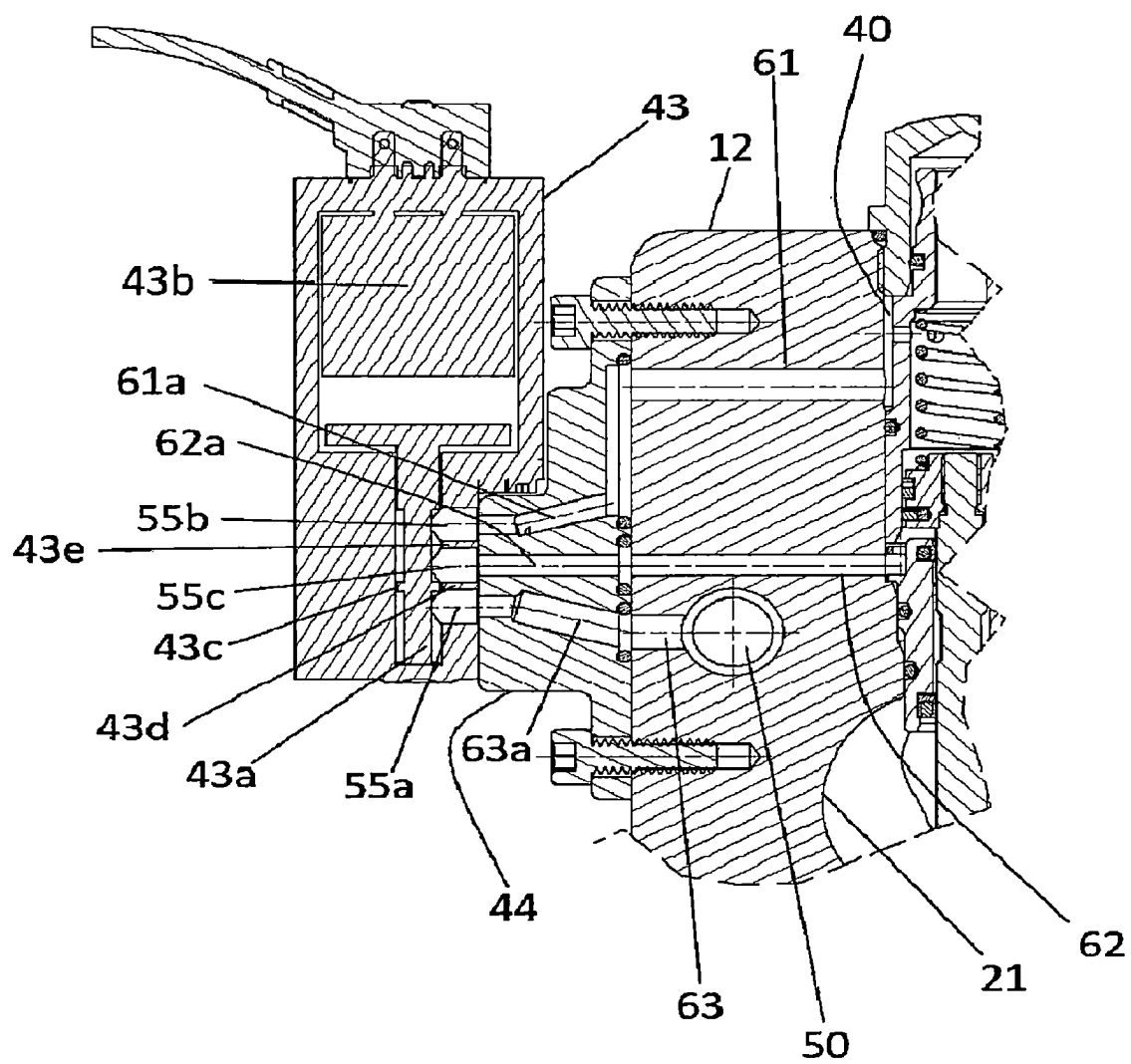


FIG. 3A1

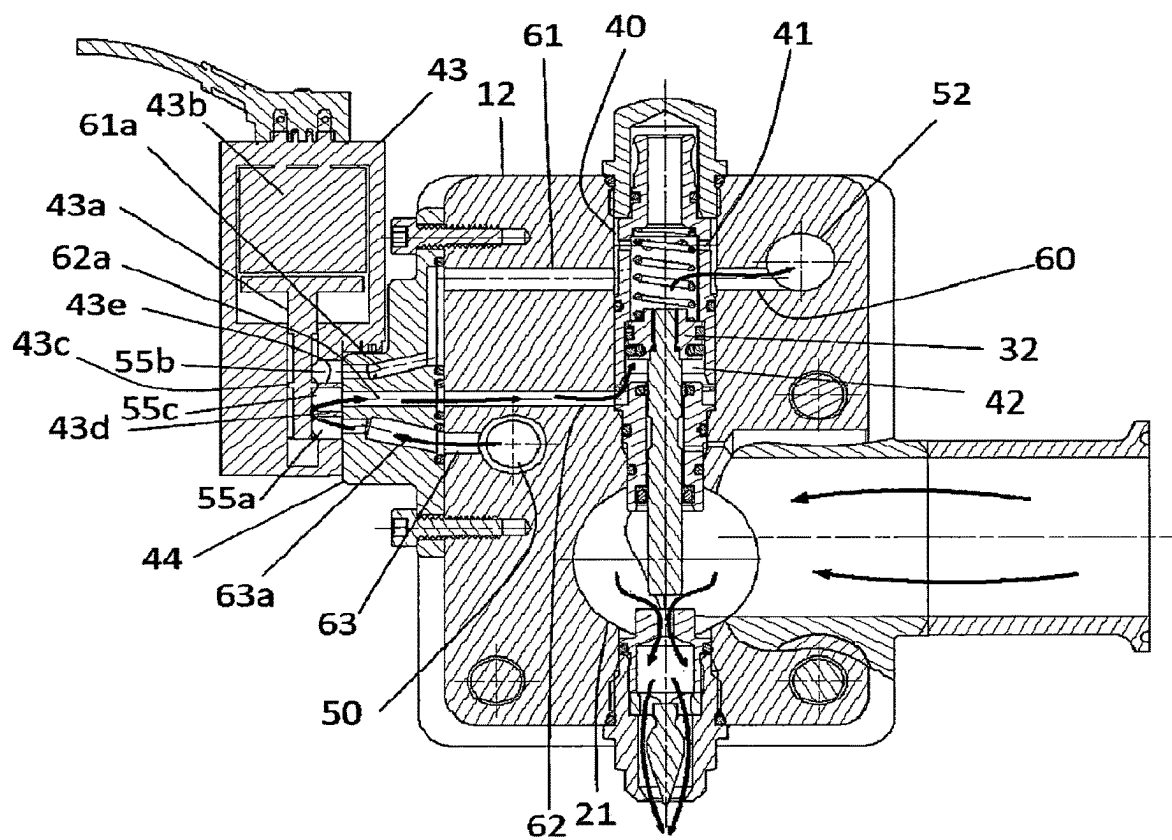


FIG. 3B

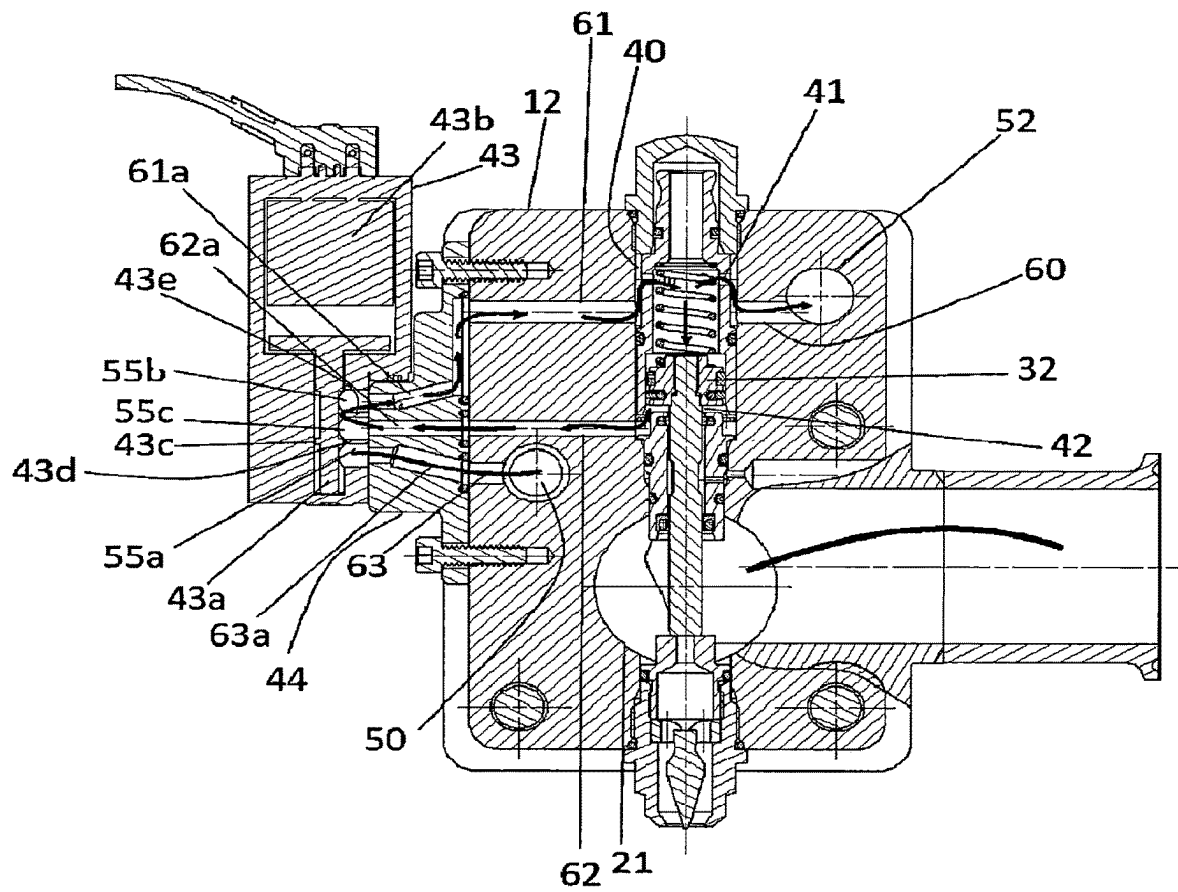


FIG. 3C

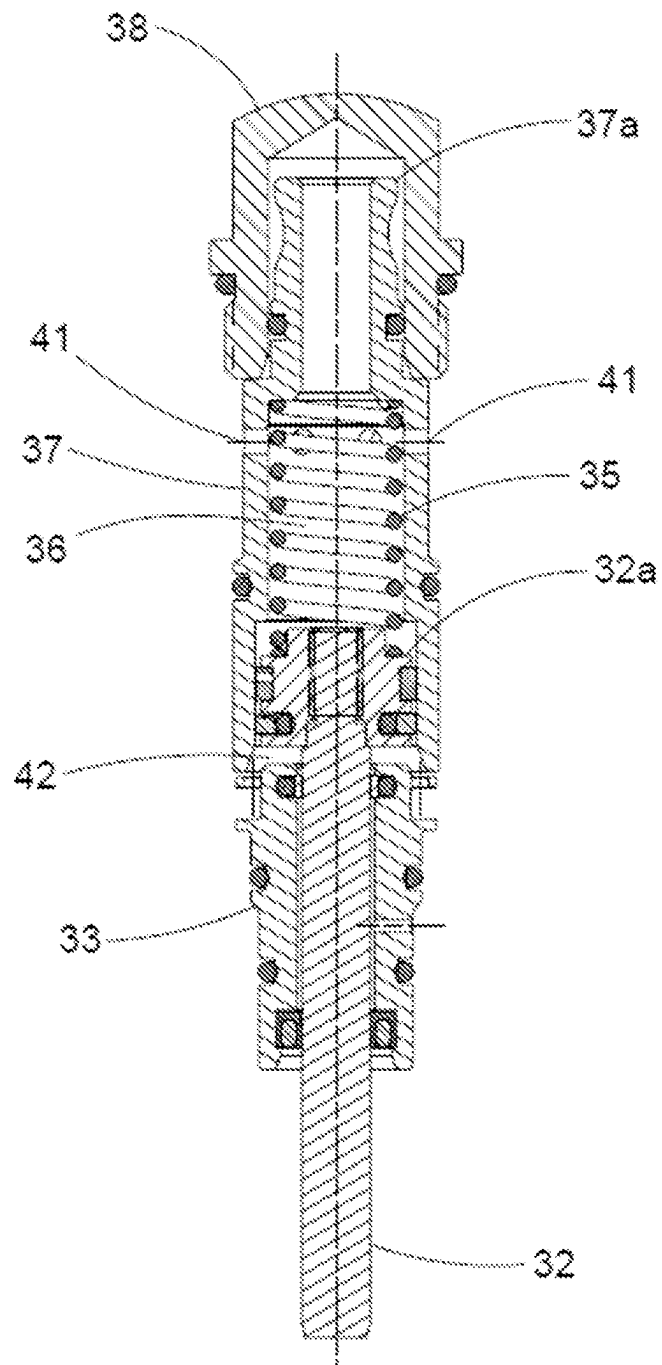


FIG. 4

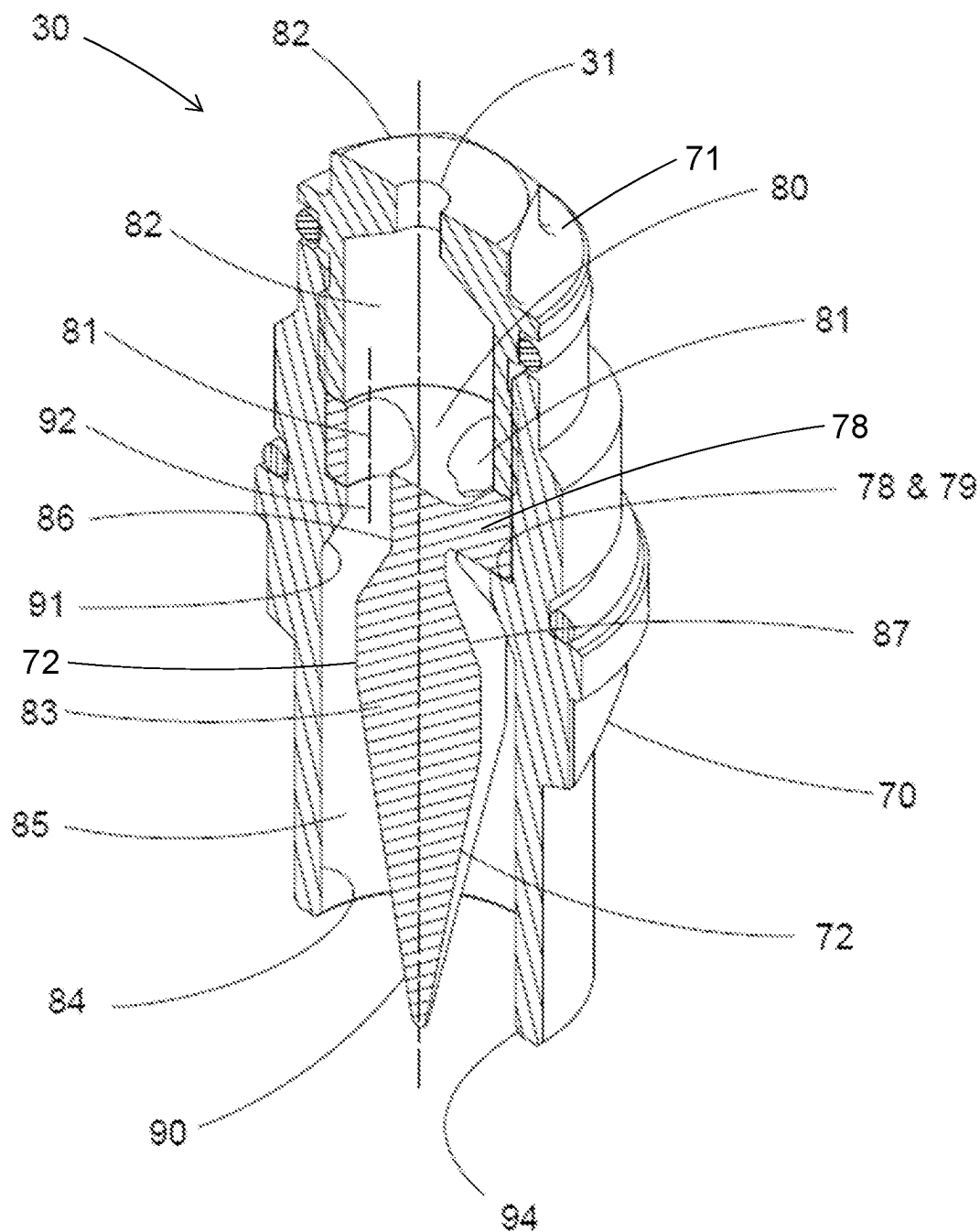


FIG. 5

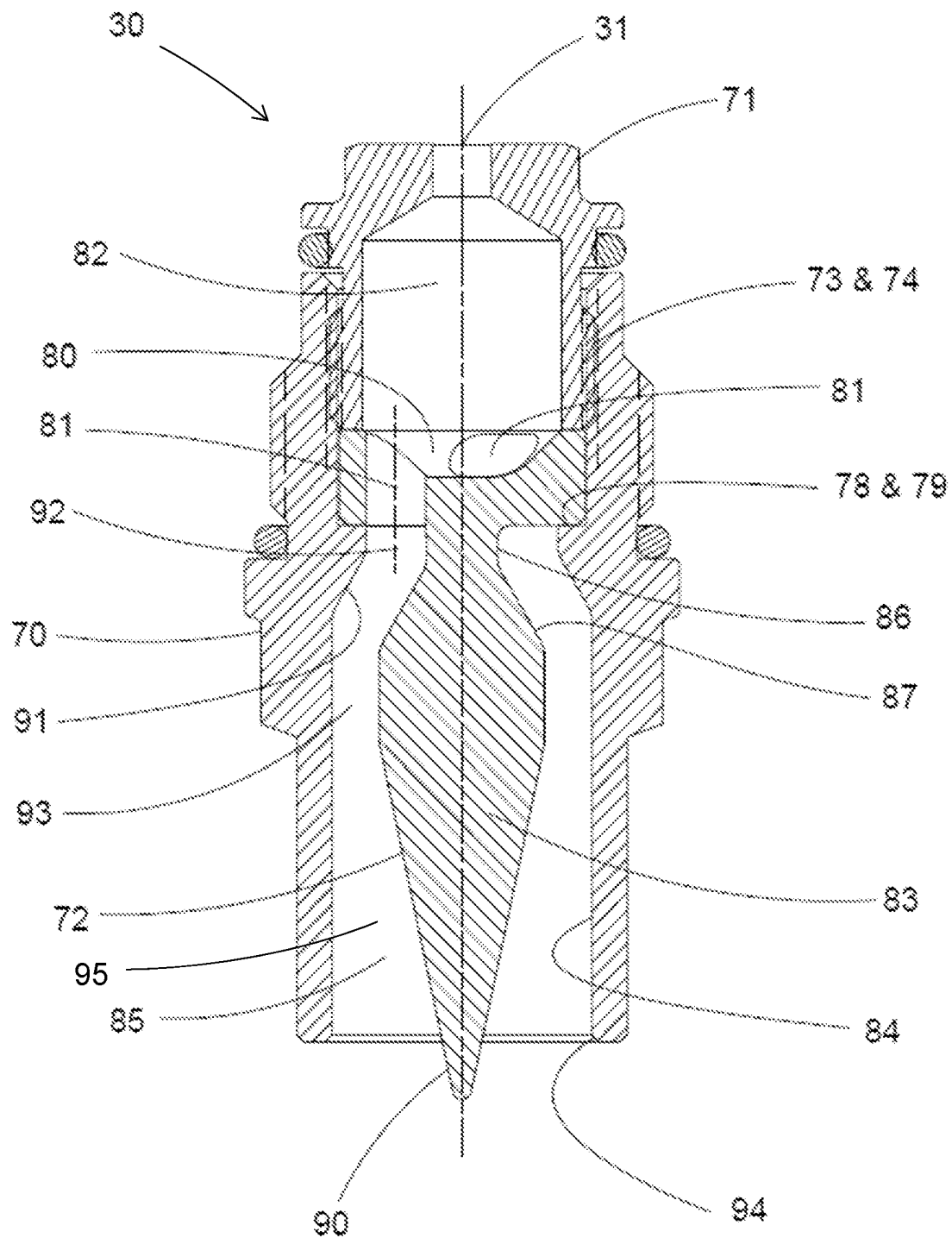


FIG. 6

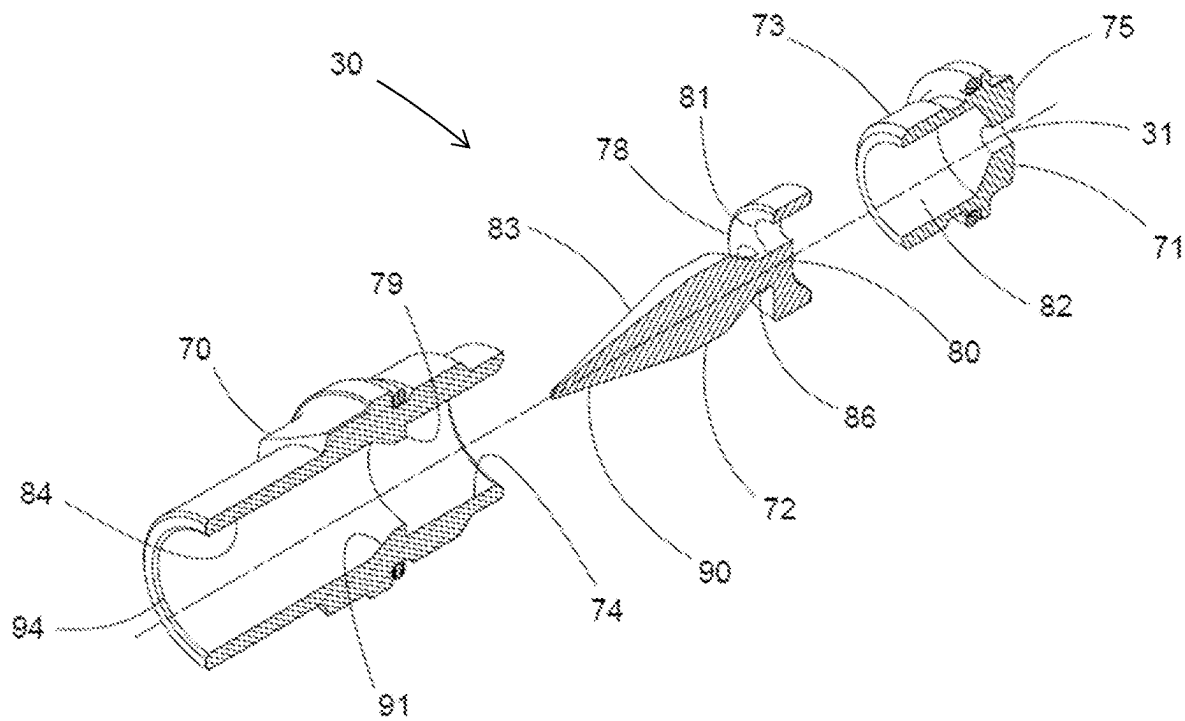


FIG. 7

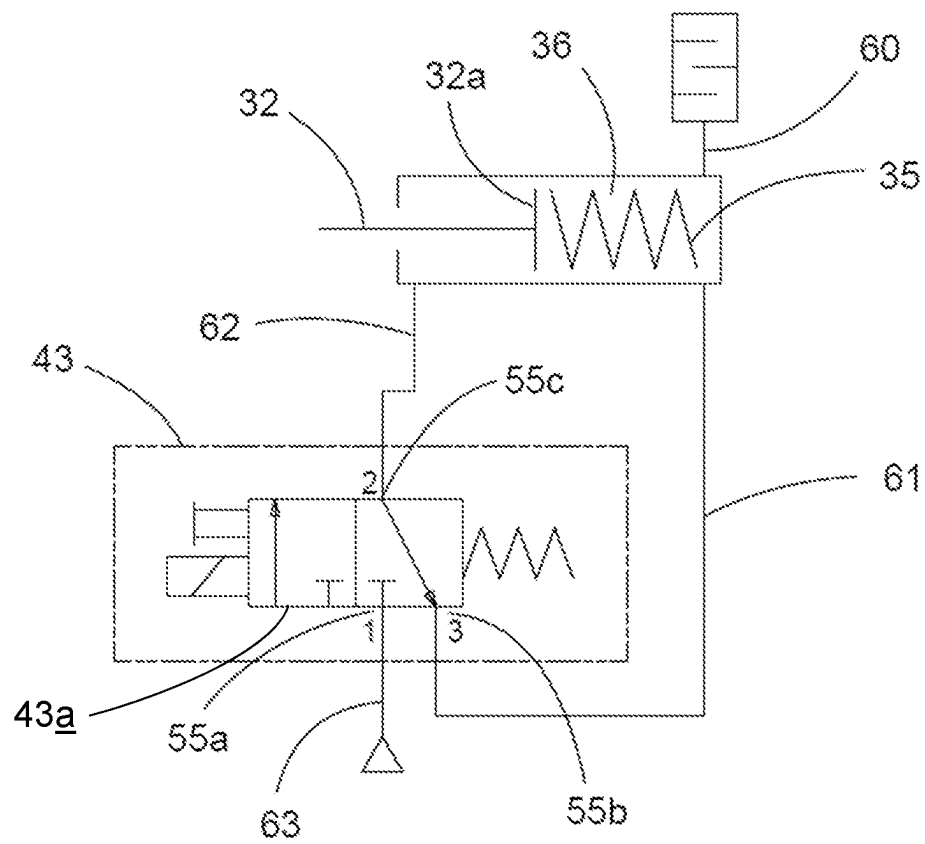


FIG. 8

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LIQUID DISPENSING SYSTEM WITH PRESSURIZED AIR OPERATED LIQUID CONTROL PISTONS

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent application claims the benefit of U.S. Provisional Patent Application No. 62/076,001, filed on Sep. 9, 2020, which is incorporated by reference.

FIELD OF THE INVENTION

The present invention relates to liquid dispensing systems, and more particularly, to liquid dispensing systems having nozzles for dispensing small controlled quantities of highly viscous liquids.

BACKGROUND OF THE INVENTION

In many industries there is a need for dispensing small, controlled quantities of highly viscous liquids. In the food industry, by way of example, in the commercial production of pizzas, it is required to dispense small droplet sized quantities of sauces onto the pizza dough. Because of the thick nature of the sauce, it is difficult to rapidly dispense closely controlled small liquid droplets as desired. Moreover, if the sauce contains solids that can clog the nozzle passages, the flow passages must be sized larger making it even more difficult to control the dispensing of small droplets and often resulting in undesirable splattering of discharging sauce. Furthermore, when the dispensing device uses an air operated liquid control piston, rapid operation of the piston is limited by the compressibility of the controlling air. Additionally, when air operated devices are spring returned, the springs return force can be limited up to roughly half of the air pressure's force used to open the device, which resists rapid piston closure.

OBJECTS AND SUMMARY OF THE INVENTION

It is an object of the invention to provide a liquid dispensing system having spray nozzles effective for dispensing precisely controlled small droplet sized quantities of highly viscous liquids.

Another object is to provide a liquid dispensing system as characterized above that is effective for rapidly depositing precisely controlled pixel sized droplets without undesirable splattering of the liquid.

A further object is to provide a liquid dispensing system of the above kind in which the spray nozzles are operable with larger inlet passages less susceptible to clogging from the solids content in the liquid.

Yet another object is to provide such a liquid dispensing system which can be selectively operated for dispensing different sized precisely controlled small droplets.

Another object is to provide such a liquid dispensing system that can be operated more rapidly.

A further object is to provide a liquid dispensing system of such type that have air actuated pistons with return springs the function of which is less resistant to air pressures used in operating the system.

Still another object is to provide a liquid dispensing system of the foregoing type that is relatively simple in design and lends itself to economical manufacture and efficient usage.

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Other objects and advantages of the invention will become apparent upon reading the following detailed description and upon reference to the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a fragmentary perspective of an illustrative modular constructed liquid dispensing system in accordance with the invention;

FIG. 2 is a perspective of the fully assembled liquid dispensing system;

FIG. 3 is a vertical section of one of the liquid dispensing modules of the illustrated liquid dispensing system;

FIGS. 3A-3C are sequential views depicting the liquid dispensing system valve stem of the solenoid control valve in its sequential operating positions;

FIG. 3A1 is an enlarged view of the solenoid control valve in the position of FIG. 3A.

FIG. 4 is an enlarged vertical section of the central liquid control piston assembly of the liquid dispensing modules shown in FIG. 3;

FIG. 5 is an enlarged perspective, in vertical section, of the spray nozzle assembly of one of the liquid dispensing modules of the illustrated system;

FIG. 6 is a vertical section of the spray nozzle assembly shown in FIG. 5;

FIG. 7 is an exploded perspective of the spray nozzle assembly shown in FIGS. 5 and 6; and

FIG. 8 is a diagrammatic depiction of a solenoid control valve associated with each respective liquid dispensing module.

While the invention is susceptible of various modifications and alternative constructions, a certain illustrative embodiment thereof has been shown in the drawings and will be described below in detail. It should be understood, however, that there is no intention to limit the invention to the specific form disclosed, but on the contrary, the intention is to cover all modifications, alternative constructions, and equivalents falling within the spirit and scope of the invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

Referring now more particularly to FIGS. 1-3 of the drawings, there is shown an illustrative liquid dispensing system 10 in accordance with the invention. The illustrated liquid dispensing system 10 is in the form of a modular valve manifold 11 comprising a plurality of individual liquid dispensing modules 12 supported and retained in sealed side by side stacked relation between end blocks 13 and 14 at opposite ends secured together by tie rods 15 and nuts 16. Each module 12 includes a module nozzle support body 20 formed with a liquid supply port 21 disposed in aligned relation with liquid supply ports 21 of the adjacent modules 12 for defining a common liquid supply passage 22 communicating between a liquid inlet port 23 in the upstream end block 13 and a liquid outlet port 24 in the downstream end block 14. Hence, liquid directed to the inlet 23 communicates through each of the stacked modules 12.

Each illustrated module 12 has a respective spray nozzle 30 mounted in depending relation to an underside of the module nozzle support body 20 having an upstream liquid inlet 31 in an upper side communicating with the liquid supply passage 22. For controlling liquid from the common liquid supply passage 22 to the spray nozzle inlet 31 of the module 12 a piston 32 is supported in each module body 20

above the spray nozzle inlet **31** for reciprocating movement between a raised inlet open position and a lowered inlet closed position.

Each piston **32** in this case is supported for selective relative movement in a carrier **33** mounted in sealed relation within a vertical opening **34** of the respective module body **20** with a downstream end of the piston **32** extending through the liquid supply passage **22** for engagement with the spray nozzle inlet **31**. For biasing the piston **32** in a lowered position closing the spray nozzle inlet **31**, a return spring **35** is disposed within a spring chamber **36** of the module body **20** in interposed relation between a head **32a** of the piston **32** and a retention sleeve **37** secured within an upper end of the body opening **34** and retained by a retention cap **38** threadably engaged within an upper end of the body opening **34**. The retention sleeve **37** in this case extends downwardly about the return spring **35** and the piston head **32a**, as best depicted in FIGS. **3** and **4**. The spring retention sleeve **37** and opening **34** of the module body **20** in this case define an annular air flow passage **40** (FIG. **3**) about the retention sleeve **37** which communicates to and through the spring chamber **36** by circumferentially offset holes **41** in the spring retention sleeve **37**. A sealed piston chamber **42** is defined between opposing axial ends of the piston head **32a** and the carrier **33** **42**-(FIG. **4**).

Pursuant to an important feature of this embodiment, each module body has a pressurized air passage system controlled by a respective valve such that pressurized air that moves the piston to an open position also augments rapid movement of the piston to closure. In the illustrated embodiment, operation of the piston **32** of each module **12** between open and closed positions is controlled by a respective solenoid valve **43**, as best depicted in FIGS. **3A-C** and **A1**. Each module solenoid valve **43** has a housing **43b** and a piston with a valve stem **43a** within a cylindrical piston chamber of the housing **43b**. The solenoid valve housing **43b** is attached to its respective module body **20** with a solenoid mounting block **44** mounted in sealed relation to its respective module body **20** by screws **45**. The module bodies **20** each have an air supply port **50** aligned with the air supply port **50** of each adjacent module body to define a common air inlet passage communicating with a system air inlet port **51** in the end block **14**. The module bodies **20** further each have an air outlet port **52** aligned to define a common outlet air passage communicating with a system exhaust outlet port **53** in the end block **14**. The air supply port **50** of the module body **20** communicates through inlet air passages **63**, **63a** in the module body **20** and solenoid mounting block **44** to an air inlet port **55a** of the solenoid valve **43**. The air outlet port **52** of the module body **20** communicates through outlet passage **60** with the return spring chamber **36** via holes **41** in the retention sleeve **37** and the annular passage **40** about the sleeve **37**, and outlet passages **61**, **61a** in the module body **20** and solenoid mounting block **44** with an exhaust port **55b** of the solenoid valve **43**. The piston chamber **42** communicates via work passages **62**, **62a** in the module body **20** and solenoid mounting block **44** with a work port **55c** of the solenoid control valve **43**.

The air inlet port **55a**, work port **55c**, and exhaust port **55b** in this case are defined by radial recesses in the solenoid housing **43b** in close axial relation to each other. The valve stem **43a** carries an annular sealing ring **43c** and is selectively movable by the solenoid valve for engaging sealing surfaces defined by respective ends of radial walls **43d** and **43e** between the ports **55a** and **55c**, and between ports **55c** and **55b** and the surrounding piston chamber. Hence, when the solenoid valve **43** is in its natural or non-energized state,

as depicted in FIG. **3A**, inlet air pressure at the solenoid valve air outlet port **55a** is blocked by the annular sealing ring **43c** contact with the radial wall **43d** and surrounding surface of the piston chamber preventing pressurized air at the air supply port **50** of the module body **20** from communicating with the piston chamber **42** via passages **62a**, **62** in the solenoid mounting block **44** and module body **20**. Additional passage routing when the solenoid valve **43** is in its natural or non-energized state connects ports **55c**, **55b** of solenoid valve **43**, allowing communication of air between piston chamber **42** and outlet port **52** in the module body **20** via passages **62**, **62a** in the module body **20** and solenoid mounting block **44**, outlet passages **61a**, **61** in the solenoid mounting block **44** and module body **20**, the annular passage **40** about through the spring **35** via holes **41**, and outlet passage **60**.

When the solenoid valve **43** is energized, the valve stem **43a** is moved from the first position in which the valve stem sealing ring **43c** contacts the first radial wall **43d** to a second position where the valve stem sealing ring **43c** contacts the second radial wall **43e**, connecting solenoid valve ports **55a**, **55c** while closing the connection of the exhaust port **55b** to atmosphere. Pressurized air at the air supply port **50** of module body **20** then communicates with pressure chamber **42** via passages **63**, **63a** in the module body **20** and solenoid mounting block **44**, solenoid valve ports **55a**, **55c** and passages **62a**, **62** causing the piston **32** to stroke upwardly, opening the nozzle inlet **31** and compressing the return spring **35** as depicted in FIG. **3B**. The upward stroke of the piston head **32a** imparts a positive air displacement within the spring chamber **36** resulting in a slight pressure increase. The pressure increase in the spring cavity **36** drains through holes **41** in the spring retention sleeve **37**, annular passage **40**, outlet passage **60**, and air outlet port **52** to atmosphere pressure (FIGS. **3** and **4**). The nozzle inlet **31** remains open, allowing liquid flow, from the common liquid supply passage **22** through the spray nozzle **30** to atmosphere while the solenoid is energized.

When un-energized, the solenoid valve **43** shifts back to the natural state as depicted in FIG. **3C**. Inlet air pressure at solenoid port **55a** is again blocked preventing pressurized air from entering the device. Rapid decompression of the pressurized air in piston chamber **42** causes a migration of elevated pressure within passages **62**, **62a** in the module body **20** and solenoid mounting block **44**, ports **55c** and **55b** of solenoid valve **43**, outlet passage **61a** and **61** in the solenoid mounting block **44** and module body **20**, spring chamber **36**, outlet passage **60**, and outlet port **52** as the system pressure in piston chamber **42** is released and equalizes with the atmosphere. Migrating pressure in the annular passage **40** communicates through holes **41** in spring retention sleeve **37** causing elevated pressure within spring chamber **36** and acting on the surface area of piston head **32a** within the spring chamber **36** resulting in a momentary downward force supplementing the constant downward force from return spring **35** opposing the decompressing pressure in the piston chamber **42**, returning piston **32** to its natural state, closing passage **31** and stopping liquid flow through spray nozzle **30**, from common liquid supply passage **22**. An appreciable decrease in time required to return the piston **32** to its natural state is attributed to the momentary increase in pressure within the spring chamber **36**. All passages and cavities downstream of solenoid valve **43** including spring chamber **36** intrinsically return to atmospheric pressure through an outlet port **52** effectively removing the supplemental force the transient pressure applied to piston head **32a**.

Further operation of the liquid dispensing module 12 by again energizing solenoid 43 is unaffected by the previous cycles increased pressure in the spring chamber 36 as the increased pressure is transient and quickly returns to atmosphere pressure allowing the increased pressure to have the desired effect on piston's 32 opening stroke without effecting the piston's 32 closing stroke. As it will become apparent, the solenoid valve 43 can be cycled at predetermined rates for the particular dispensing operation with the piston's 32 variable open time providing a varying pixel volume.

In accordance with a further aspect of the present embodiment, each spray nozzle module 12 is operative for dispensing controlled small round pixel sized droplets of highly viscous liquid as an incident to cycling of the piston 32 even when the liquid has an appreciable solids content. Each spray nozzle 30 with particular reference to FIGS. 5, 6, and 7, in this case comprises a nozzle body 70, a nozzle seat 71 and an internal nozzle core 72. The nozzle seat 71 in this instance has an externally threaded cylindrical downstream end 73 that is threadedly engaged within an upstream cylindrical end 74 of the nozzle body 70 to secure the nozzle core 92 within the nozzle body 70. An upstream end 75 of the nozzle seat 71 defines the predetermined size liquid inlet 31, which is at the upstream end of the assembly. The nozzle core 72 in this instance has an upstream cylindrical mounting flange 78 positioned on an annular ledge 79 within the nozzle body 70 and retained in place by the nozzle seat 71, although it will be appreciated that other methods may be used to secure the nozzle core 72 within the nozzle body 70.

The cylindrical mounting flange 78 of the core 72 has a concavely configured (relative to the direction of fluid flow) downstream end wall 80 formed with a plurality of circumferentially spaced axially oriented liquid orifices 81. These liquid orifices that communicate between an expansion cavity 82 of the nozzle seat 71 and an annularly configured liquid discharge passage defined between the nozzle core 72 and the nozzle body 70 for directing liquid in a controlled fashion for optimum dispensing in small droplet form as will become apparent. It will be understood that while the illustrated nozzle 30 comprises a multi-part assembly, alternatively, it could have a one-piece construction or fewer or greater assembled parts.

In carrying out this aspect of the present embodiment, the nozzle core 72 has a teardrop shaped pintle 83 which together with the internal circumferential surface of the surrounding nozzle body 70 defines an expanding discharge passage 85 that reduces exit velocity of the dispensed liquid for maintaining a desired flow rate and consistent droplet size of the highly viscous discharging liquid. To that end, the illustrated pintle 83 (see FIGS. 5, 6, and 7) has a relatively small diameter upstream end section 86 extending centrally from the mounting flange 78, a radially outwardly extending curved section 87 adjacent to the upstream end, and an inwardly tapered, relatively long conical terminal end section 90. As noted, the nozzle body 70 has a generally hollow cylindrical configuration with the internal circumferential surface of the nozzle body 70 defining the outer wall of the annular discharge passage 85 about the core section 72. The inner wall of the discharge passage 85 is defined by the outer surface of the pintle 83. In this case, the internal circumferential surface of the nozzle body 70 includes a radially outwardly directed section 91 that extends in surrounding relation to the outwardly curved section 87 of the nozzle core 72 and a uniform diameter section 84 that then extends downstream substantially the remaining length of the pintle 83. The design is unique in that flow through the annular discharge passage effects inward expansion of viscous liquid

during travel through the nozzle body. The geometry of the pintle defines the inner diametric wall of the annular flow path while providing a structure against which a vacuum due to the flow expansion can be formed. The deceleration of the liquid within the expanding annular discharge passage is a function of the surface tension and the capillary forces' ability to draw a vacuum and resist flow.

In operation, with continued reference to FIG. 3, when the piston 32 is in a raised inlet open position, liquid is permitted to pass through the nozzle inlet 31 in a controlled fashion into the expansion cavity 82 defined within the cylindrical downstream end of the nozzle seat 71. Liquid passing through the nozzle inlet 31 is directed against an impingement surface defined by the concave downstream end wall 80 of the expansion cavity 82. This causes liquid to fill the expansion cavity 82 and then subsequently extrude from the expansion cavity through the series of circumferentially spaced orifices 81 into the discharge passage 85. Moreover, the size of each of the orifices 81 is at least as large as the nozzle inlet 31 to allow solid particles in the liquid to flow from the expansion cavity 82 to the fluid discharge passage 85 without clogging. The collective area of the circumferentially spaced orifices 81 is greater than the area of the nozzle inlet 31 such that the velocity of the liquid passing through the orifices 81 is inversely proportional to the ratio of the size of the orifices 81 to the size of the nozzle inlet 31.

More specifically, the circumferential orifices 81 at the downstream end of the expansion cavity 82 communicate with an inlet section 92 of the discharge passage 85 that is defined between the outwardly directed wall section 91 of the nozzle body 70 and the pintle 83 of the nozzle core 72. The cross-sectional area of the annular inlet section 92 may increase as the section extends in the downstream direction such that the velocity of the fluid in this region continues to be reduced as the cross-sectional area of the discharge passage expands. A slight reduction in the cross-sectional area of the discharge passage 85 in a subsequent stabilizing section 93 of the discharge passage 85 (again defined by the outer surface of the pintle 83 and the inner circumferential surface of the nozzle body 70) immediately downstream of the inlet section 92 can provide a slight increase in pressure. This increase in pressure stabilizes and balances the flow removing individual jet streams caused by the fluid entering the inlet section 92 of the discharge passage 85 through the series of orifices 81 and allows uniform flow along the internal wall surface of the nozzle body 70. The cross-sectional area of the stabilizing section 93 remains constant through this region as the fluid gains stability.

Downstream of the stabilizing section 93, the liquid enters a final expansion section 95 defined by the inwardly tapered terminal end section 90 of the nozzle core 72 that extends downstream to a nozzle mouth 94 defined at the downstream end of the nozzle body 70. The progressively increasing cross-sectional area of the final expansion section 95 is achieved through the reducing conical diameter of the pintle 83 in the terminal end section 90 while the inner circumferential surface of the nozzle body 70 is maintained at a consistent diameter. The pintle 83 helps stabilize the fluid and enables greater expansion of the liquid than could be achieved with nozzle core having a simple uniform diameter. Sustained contact of the liquid with inner and outer wall surfaces of the discharge passage is a function of the surface tension of the liquid.

The cross-sectional area of the final expansion section 95 at the nozzle mouth 94 defines the exit velocity of the liquid, which is inversely proportional to the cross-sectional area at the nozzle mouth 94 in relation to the area of nozzle inlet 31.

The terminal end section **90** of the nozzle core **72** preferably extends slightly beyond the nozzle mouth **94** to assist in breaking the surface tension of the liquid with the inner circumferential surface of the nozzle body **70** without impacting the outer diameter of the discharging liquid stream. Having the inner circumferential surface of the nozzle body **70** at a constant diameter helps establish a consistent diameter of the boundary layer of the liquid when it exits the nozzle, which assists in maintaining the desired droplet diameter independent of the distance of the nozzle from the target.

It has been found that a dramatic reduction in the velocity of the liquid can be achieved through progressively increasing the cross-sectional area of the discharge passage **85**. The inward expansion of the discharge passage **85** is achieved by progressively reducing the diameter of the pintle **83** while maintaining the inner circumferential surface of the nozzle body **70**. This helps produce discharging liquid with a consistent stream diameter. The reduced velocity of the liquid allows it to be dispensed without splattering. This further allows the utilization of larger nozzle inlet orifices **31** for enabling the dispensing of liquids with larger solids content. Once the discharge passage of the nozzle is initially filled with viscous fluid, the surface tension of the liquid will keep the nozzle **30** charged with liquid ready to be dispensed upon opening of the nozzle inlet **31**. Because the liquid can be substantially incompressible, an exact relationship can be maintained between the liquid volume entering the nozzle **30** through the inlet **31** and the liquid exiting the nozzle mouth **94**. Cycling the piston **32** to open and close the inlet orifice **31** at a rapid rate, such as 50 milliseconds, has been found to produce small, consistent droplets of liquid that are discharged at a reduced exit velocity. This allows the discharging droplets to be deposited on a target, such as a target about 2 inches from the nozzle, without splattering.

From the foregoing, it can be seen that a liquid dispensing system is provided that can be selectively operated for dispensing precisely controlled droplets without undesirable splattering of the dispensed liquid. The system further includes spray nozzles that are less susceptible to clogging from the solids content of the liquid. Yet the liquid dispensing system and the liquid spray nozzles thereof are relatively simple in design and lends itself to economical manufacture and efficient usage. While the spray nozzles have been shown and described in conjunction with an illustrative liquid dispensing system, it will be understood that the spray nozzles could be used in other applications for dispensing controlled relatively small quantities of liquid.

What is claimed:

1. A spray nozzle assembly comprising:

- a nozzle support body (**20**) having a liquid supply port (**22**) for coupling to a liquid supply, said nozzle support body (**20**) further having an air supply port (**50**) for coupling to a pressurized air supply and air outlet port (**52**);
- a spray nozzle (**30**) supported by said nozzle support body (**20**);
- a piston (**32**) having a piston head (**32a**) supported for movement in an opening (**34**) of said nozzle support body (**20**) with a downstream end of the piston (**32**) being selectively engageable with a spray nozzle liquid supply inlet (**31**) for controlling the flow of liquid to and discharge from said spray nozzle (**30**);
- a return spring (**35**) supported within a spring chamber (**36**) in said nozzle support body opening (**34**) on an

upstream side of said piston head (**32a**) for biasing said piston (**32**) to a spray nozzle liquid supply inlet closing position;

said nozzle body opening (**34**) defining a piston chamber (**42**) on a downstream side of said piston head (**32a**) opposite of said spring chamber (**36**);

said nozzle support body (**20**) having an air passage system communicating with a control valve (**43**),

said control valve (**43**) being selectively operable (1) for enabling communication of pressurized air from said nozzle support body air supply port (**50**) to said piston chamber (**42**) for moving said piston (**32**) to a spray nozzle liquid supply inlet open position against the biasing force of said return spring (**35**) for enabling the flow of liquid to and discharge from said spray nozzle (**30**) and (2) for preventing the flow of pressurized air through said air passage system from said nozzle support body air supply port (**50**) to said piston chamber (**42**) while venting air pressure in the said piston chamber (**42**) to atmosphere through said spring chamber (**36**) to a nozzle support body air outlet port (**52**) for causing an elevated pressure within spring chamber (**36**) acting on the piston head (**32a**) resulting in a momentary force supplementing the biasing force of said return spring (**35**) in moving said piston (**32**) to said spray nozzle liquid supply inlet closing position.

2. The spray nozzle assembly of claim 1 in which said control valve (**43**) has an air inlet port (**55a**), a work port (**55c**), and an exhaust port (**55b**) and an actuating mechanism in the form of a valve stem (**43a**) movable between (1) a first position that opens said valve inlet port (**55a**) for enabling communication of pressurized air from said nozzle support body air supply port (**50**) through said valve inlet port (**55a**) and work port (**55c**) to said piston chamber (**42**) for moving said piston (**32**) to said open position and (2) a second position that closes said valve inlet port (**55a**) blocking communication of pressurized air from said nozzle support body air supply port (**50**) through said control valve inlet port (**55a**) while opening said exhaust port (**55b**) for enabling communication of pressurized air from said piston chamber **42** through said valve work port (**55c**) and exhaust port (**55b**) to and through said spring chamber (**36**) to nozzle support body air outlet port (**52**).

3. The spray nozzle assembly of claim 1 in which said nozzle body air passage system includes a first air passage (**63**) for communicating pressurized air from said nozzle support body air supply port (**50**) to said control valve (**43**), a work passage (**62**) for communicating pressurized air from said control valve (**43**) to said piston chamber (**42**), and outlet air passage (**61**) for communicating pressurized air from said control valve (**43**) to and through said piston chamber (**42**) to said nozzle support body air outlet port (**52**).

4. The spray nozzle assembly of claim 3 in which said control valve (**43**) has a valve stem (**43a**); and said control valve (**43**) is operable for moving said valve stem (**43a**) between (1) a first position that enables communication of pressurized air from said air supply port (**50**) through said nozzle support body first air passage (**63**) and work passage (**62**) to said piston chamber (**42**) for moving said piston (**32**) against the force of said return spring (**35**) to said open position while blocking communication of pressurized air to said nozzle support body outlet air passage (**61**), spring chamber (**36**) and nozzle support body air outlet port (**52**), and (2) a second position that blocks communication of pressurized air from said first air passage (**63**) to said work passage (**62**) while enabling communication of pressurized air from said piston chamber (**42**) through said work passage

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(62), outlet air passage (61) to and through said spring chamber (36) and to said nozzle support body air outlet port (52).

5 5. The spray nozzle assembly of claim 4 in which said return spring (35) is disposed within a retention sleeve (37), and when said valve actuating mechanism (43) is said second position pressurized air communicates from said piston chamber (42) about and through said retention sleeve (40) to said nozzle body air outlet port (52).

10 6. The spray nozzle assembly of claim 1 in which said control valve is a solenoid valve.

7. The spray nozzle assembly of claim 1 in which said piston (32) is supported within a piston carrier (33) mounted in sealed relation within said nozzle support body opening (34); and said piston carrier (33) and piston head (32a) 15 define the limits of said piston chamber (42) within said nozzle body opening (34).

8. The spray nozzle assembly of claim 1 in which said spray nozzle liquid supply inlet is an inlet (31) in an upstream end of said spray nozzle (30). 20

9. The spray nozzle assembly of claim 1 including a plurality of said nozzle support bodies each having a respective one of said control valves (43) and spray nozzles (30).

10. The spray nozzle assembly of claim 9 in which said plurality of said nozzle support bodies (20) are disposed in side by side relation to each other with the liquid supply port 25 (22), air supply port (50), and air outlet port (52) of each

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nozzle support body (20) being in fluid communication with the respective liquid supply port (22), air supply port (50) and air outlet port (52) of the other of the plurality of nozzle support bodies.

11. The spray nozzle assembly of claim 1 in which said spray nozzle includes a nozzle body having a hollow cylindrical configuration defining an internal circumferential surface; and an internal nozzle core arranged within the nozzle body and including a teardrop shaped pintle having an upstream end section, a radially outward curved section adjacent to the upstream end section and a radially inwardly tapered conical terminal end section, and an annular discharge passage being defined between an outer surface of the pintle and the internal circumferential surface of the nozzle body.

12. The spray nozzle assembly of claim 11 in which said spray nozzle includes a nozzle seat having said liquid supply inlet and an expansion cavity in fluid communication with the liquid inlet, the expansion cavity terminating in a downstream end wall, and said annular discharge passage being in fluid communication with said expansion cavity.

13. A modular manifold spraying system comprising a plurality of spray nozzle assemblies as defined in claim 1 supported in sealed side by side stacked relation to each other.

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